

Decision Checklist: Supervisor Meeting

Purpose: Capture all critical decisions during the meeting to avoid ambiguity later

Instructions: Check boxes and fill in answers as decisions are made

1. Scope Freeze & Success Criteria

1.1 Geometry Set

- ☐ **Current geometry set is sufficient for thesis**
 - L-bracket (2 load cases)
 - Tapered plate (1 load case)
 - Ribbed channel (2 load cases)
 - **Decision:** ☐ Yes ☐ No
 - **Notes:** _____
- ☐ **Benchmark 2 (motor housing) is no longer required**
 - **Decision:** ☐ Yes ☐ No
 - **Notes:** _____
- ☐ **GA baseline is no longer required**
 - **Decision:** ☐ Yes ☐ No
 - **Notes:** _____

1.2 Ansys Validation Requirements

- ☐ **Number of layouts to validate:**
 - **Decision:** _____ layouts (suggested: 3-5)
 - **Which layouts:**
 - ☐ Best diff-FEA layout (L-bracket)
 - ☐ Best PINN layout (L-bracket)
 - ☐ Best MMC layout (L-bracket, each load)
 - ☐ At least one non-L-bracket geometry (tapered or ribbed)
 - ☐ Other: _____
 - ☐ **Error threshold acceptable:**
 - **Decision:** < _____ % error (suggested: <20%)
 - **Notes:** _____
 - ☐ **Validation is sufficient if error < threshold:**
 - **Decision:** ☐ Yes ☐ No
 - **Notes:** _____
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2. 3D Work Decision

2.1 3D Requirement

- ☐ **Is 3D benchmark required for passing?**
 - **Decision:** ☐ Required ☐ Optional ☐ Not required
 - **Notes:** _____

2.2 If 3D Required

- ☐ **Is PINN-only 3D acceptable?**
 - (MMC remains 2D-only, as 3D MMC is out of scope)
 - **Decision:** ☐ Yes ☐ No
 - **Notes:** _____
- ☐ **Which 3D geometry preferred?**
 - **Decision:** ☐ Extruded L-bracket ☐ Simplified motor-housing ☐ Other: _____
 - **Notes:** _____
- ☐ **Dataset size for 3D:**
 - **Decision:** _____ samples (suggested: 50-100)
 - **Notes:** _____
- ☐ **Timeline for 3D work:**
 - **Decision:** _____ weeks allocated
 - **Notes:** _____

2.3 2D Work Status

- ☐ **Current 2D multi-geometry results are sufficient as core comparative study**
 - (Regardless of 3D decision)
 - **Decision:** ☐ Yes ☐ No
 - **Notes:** _____
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3. Grading Emphasis & Thesis Balance

3.1 Thesis Emphasis

- ☐ **How should final thesis be balanced?**
 - **Decision:**
 - ☐ More weight on comparative discussion and generalization (current strong numbers)
 - ☐ More weight on deep derivations
 - ☐ More weight on additional experiments
 - ☐ Balanced across all three
 - **Notes:** _____

3.2 MMC Normalization

- ☐ **MMC normalized values on ribbed-channel:**
 - **Decision:**

- ☐ Acceptable if clearly explained (scaling and trends)
 - ☐ Must be fixed/renormalized before thesis submission
 - **Notes:** _____
 - ☐ **MMC normalization to Joules:**
 - **Timeline:**
 - ☐ Must be done before writing begins
 - ☐ Can be done during writing phase
 - ☐ Can be explained in text without conversion
 - **Notes:** _____
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4. Writing Expectations & Timeline

4.1 Chapter Delivery Schedule

- ☐ **Chapters 1-3 draft (Introduction, Literature Review, Theoretical Framework):**
 - **Target date:** _____ weeks from meeting
 - **Decision:** ☐ Approved ☐ Needs adjustment
 - **Notes:** _____
- ☐ **Chapters 4-5 draft (Methodology, Results):**
 - **Target date:** _____ weeks from meeting
 - **Decision:** ☐ Approved ☐ Needs adjustment
 - **Notes:** _____
- ☐ **Full draft (all chapters):**
 - **Target date:** _____ weeks from meeting
 - **Latest possible date:** _____ (back-solve from defense date)
 - **Decision:** ☐ Approved ☐ Needs adjustment
 - **Notes:** _____

4.2 Review Process

- ☐ **Number of feedback rounds supervisor is willing to do:**
 - **Decision:** _____ rounds (suggested: 2-3)
 - **Notes:** _____
 - ☐ **Review turnaround time:**
 - **Decision:** _____ weeks per round
 - **Notes:** _____
 - ☐ **Communication method:**
 - **Decision:** ☐ Email ☐ In-person ☐ Virtual meeting ☐ Other: _____
 - **Notes:** _____
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5. Final Thesis Scope Statement

5.1 Approved Scope

- ☐ **One-sentence thesis scope statement:**

▸ **Decision:**

[Fill in approved statement]

▸ **Notes:** _____

Example options:

- "Comparative study of multi-geometry PINN vs MMC on three 2D benchmark assemblies with Ansys validation on selected layouts"
 - "Comparative study of multi-geometry PINN vs MMC on three 2D benchmark assemblies, with 3D PINN extension, and Ansys validation on selected layouts"
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6. Final Tasks Checklist (Sufficient to Pass)

6.1 Technical Tasks

- ☐ **Run and document Ansys validation on N layouts**
 - N = _____ (from section 1.2)
 - **Timeline:** _____ weeks
 - **Decision:** ☐ Approved ☐ Needs adjustment
- ☐ **Normalize MMC compliances and regenerate unified comparison plots**
 - **Timeline:** _____ weeks
 - **Decision:** ☐ Approved ☐ Needs adjustment
- ☐ **Produce dataset coverage figure for 1020-sample corpus**
 - **Timeline:** _____ weeks
 - **Decision:** ☐ Approved ☐ Needs adjustment
- ☐ **Additional technical tasks (if any):**
 - _____
 - **Timeline:** _____ weeks
 - **Decision:** ☐ Approved ☐ Needs adjustment

6.2 Writing Tasks

- ☐ **Deliver full thesis draft by date:**
 - **Date:** _____
 - **Decision:** ☐ Approved ☐ Needs adjustment
 - ☐ **Additional writing requirements:**
 - _____
 - **Decision:** ☐ Approved ☐ Needs adjustment
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7. Stop Adding Experiments

7.1 Scope Freeze Confirmation

- ☐ **Confirm:** stop adding new experiments, focus only on checklist items and writing
 - **Decision:** ☐ Approved ☐ Exceptions noted below
 - **Notes:** _____

7.2 Exceptions (if any)

- ☐ **Exception 1:**
 - **Description:** _____
 - **Justification:** _____
 - **Decision:** ☐ Approved ☐ Rejected
 - ☐ **Exception 2:**
 - **Description:** _____
 - **Justification:** _____
 - **Decision:** ☐ Approved ☐ Rejected
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8. Additional Decisions & Notes

8.1 Open Questions

- **Question 1:** _____
 - **Answer:** _____
- **Question 2:** _____
 - **Answer:** _____

8.2 Supervisor Feedback

- **Feedback 1:** _____
- **Feedback 2:** _____
- **Feedback 3:** _____

8.3 Next Steps

- **Next meeting date:** _____
 - **Communication channel:** _____
 - **Immediate action items:**
 - ☐ _____
 - ☐ _____
 - ☐ _____
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Meeting Summary

Key Decisions Made:

1. _____
2. _____
3. _____

Approved Scope: _____

Timeline: _____

Next Meeting: _____

Meeting Date: _____

Attendees: Diovandi Basheera Putra, Dr. Leonard P. Rusli

Completed by: _____

Date completed: _____